Project: PulseForge

Category: healthcare innovation

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Participation type: individual submission

Executive Summary

Healthcare is shifting from reactive treatment toward preventive and proactive care, where continuous health awareness and early detection play a central role in improving patient outcomes. This vision aligns with Egypt Vision 2030, which emphasizes digital transformation, smart healthcare technologies, and improving the quality and accessibility of healthcare services. Achieving this transformation requires intelligent monitoring systems capable of providing timely clinical insights before medical conditions progress into life-threatening emergencies.

To support this vision, PulseForge introduces a wearable cardiac monitoring platform designed to bridge the gap between clinical-grade ECG systems and consumer wearable devices. The project balances diagnostic accuracy, disease specificity, continuous monitoring, affordability, and low power consumption, making long-term cardiac monitoring more practical for high-risk patients.

The innovation is built on two key technological advances. The first is a **dynamic four-electrode configuration**, where two wrist electrodes provide continuous baseline monitoring while two chest electrodes are strategically positioned according to the patient's clinical profile and suspected infarction or any risk-facing region. This personalized placement increases the likelihood of capturing subtle ischemic electrical changes before severe symptoms develop.

By combining personalized electrode positioning with optimized signal acquisition, PulseForge provides a practical, affordable, and proactive cardiac monitoring solution that supports continuous monitoring without compromising clinical relevance. The system is designed to enable earlier recognition of high-risk cardiac abnormalities, facilitate timely medical intervention, and contribute to reducing preventable cardiovascular complications while advancing the adoption of intelligent preventive healthcare.

Problem statement

Cardiovascular diseases (CVDs) remain the leading cause of death worldwide, accounting for nearly 20 million deaths annually. Among them, Acute Myocardial Infarction (AMI) is one of the most time-critical emergencies, where every minute of delayed diagnosis increases myocardial damage, worsens clinical outcomes, and significantly raises mortality risk. Studies consistently demonstrate that reducing the time between symptom onset and medical intervention is one of the strongest predictors of patient survival.

Consequently, continuous cardiac monitoring has become a major healthcare priority, particularly for individuals at high cardiovascular risk. Continuous surveillance enables earlier recognition of ischemic changes, facilitates rapid medical intervention, and supports preventive healthcare strategies before irreversible myocardial injury occurs.

Although numerous wearable cardiac monitoring technologies have been developed, none simultaneously achieve high diagnostic accuracy, disease-specific monitoring, patient comfort, affordability, and long-term usability. This creates an unmet clinical need for a monitoring system capable of delivering clinically meaningful ECG information in a practical wearable form.

Existing Solutions & Current Gaps:

Current technologies reveal a common trade-off between diagnostic performance and wearability. Systems offering high clinical accuracy often sacrifice portability and patient comfort, whereas highly wearable solutions typically compromise signal quality, anatomical coverage, and disease specificity.

Furthermore, most wearable ECG systems are designed for general rhythm monitoring rather than targeted surveillance of Acute Myocardial Infarction, limiting their ability to detect subtle ischemic changes during the earliest stages of disease progression.

Existing Solution	**Strength**	**Limitation**
Hospital 12-lead ECG	Gold standard	Not wearable, intermittent assessment, requires clinical setting
Holter Monitor	Long recording	Bulky, passive recording, limited real-time decision support
Wireless Wearable ECG Patches	Continuous monitoring	Limited lead configuration, reduced spatial cardiac information
Smartwatches (Single/2-lead ECG)	Comfortable	Low sensitivity for ischemic detection, limited anatomical coverage
Existing Multi-electrode Wearables	Better ECG quality	Increased complexity, poor comfort, high power consumption, higher cost

Table (1)

Solution

PulseForge is a personalized wearable ECG monitoring platform designed to bridge the gap between conventional clinical ECG systems and consumer wearable devices. Instead of relying on a fixed electrode configuration, the system adopts a patient-specific monitoring strategy, where electrode placement is optimized according to the individual's clinical profile.

The system consists of four electrodes with complementary roles.

The first two electrodes remain permanently positioned on clavicular site, continuously acquiring a Lead I bipolar ECG. This provides uninterrupted baseline monitoring of heart rhythm and preserves the convenience and mobility offered by current smartwatch-based ECG systems.

The remaining two electrodes are not fixed. Instead, their placement is personalized according to the patient's clinical history, cardiovascular risk factors, presenting symptoms, baseline ECG findings, and physician assessment. They are positioned over the myocardial region considered most vulnerable to ischemia or infarction, enabling localized monitoring of electrical activity from the area at greatest clinical risk.

Rather than attempting to reproduce a complete 12-lead ECG, PulseForge concentrates its sensing capability on the myocardial territory most likely to undergo pathological changes. This targeted strategy increases the probability of detecting localized ischemic electrical alterations while maintaining a lightweight, wearable, and energy-efficient architecture.

To support high-quality signal acquisition, the proposed electrode design incorporates advanced conductive nanocomposite materials that enhance skin-electrode contact, reduce motion artifacts, and improve signal stability. These materials are selected based on established evidence in the scientific literature to maximize recording performance without adding unnecessary system complexity.

The acquired ECG signals are processed using a hybrid acquisition principle combining a conventional bipolar Lead I with customized localized chest leads

referenced to a software-generated Modified Central Reference (MCR). This enables the device to monitor patient-specific myocardial regions while preserving a simple four-electrode configuration.

By integrating personalized electrode positioning, targeted myocardial monitoring, and optimized signal acquisition, PulseForge provides a practical and clinically focused wearable solution for continuous surveillance of high-risk patients, supporting earlier recognition of cardiac deterioration and facilitating timely medical intervention.

Technical Design & Prototype Components

1. ECG Acquisition Module

Purpose: Acquire high-quality cardiac electrical signals.

Components:

- Personalized 4-electrode ECG array
- Recommended Proposed conductive electrode interface (Graphene–PEDOT microneedle design)
- AD8232 Analog Front-End

2. Physiological Monitoring Module

Components:

- MAX30102
- Heart Rate
- SpO_2

3. Processing Unit

Components:

- ESP32
- Signal Filtering
- Feature Extraction
- ST Analysis
- Sensor Fusion

4. Clinical Decision Module

- Risk Assessment
- Patient History
- Electrode Selection Logic

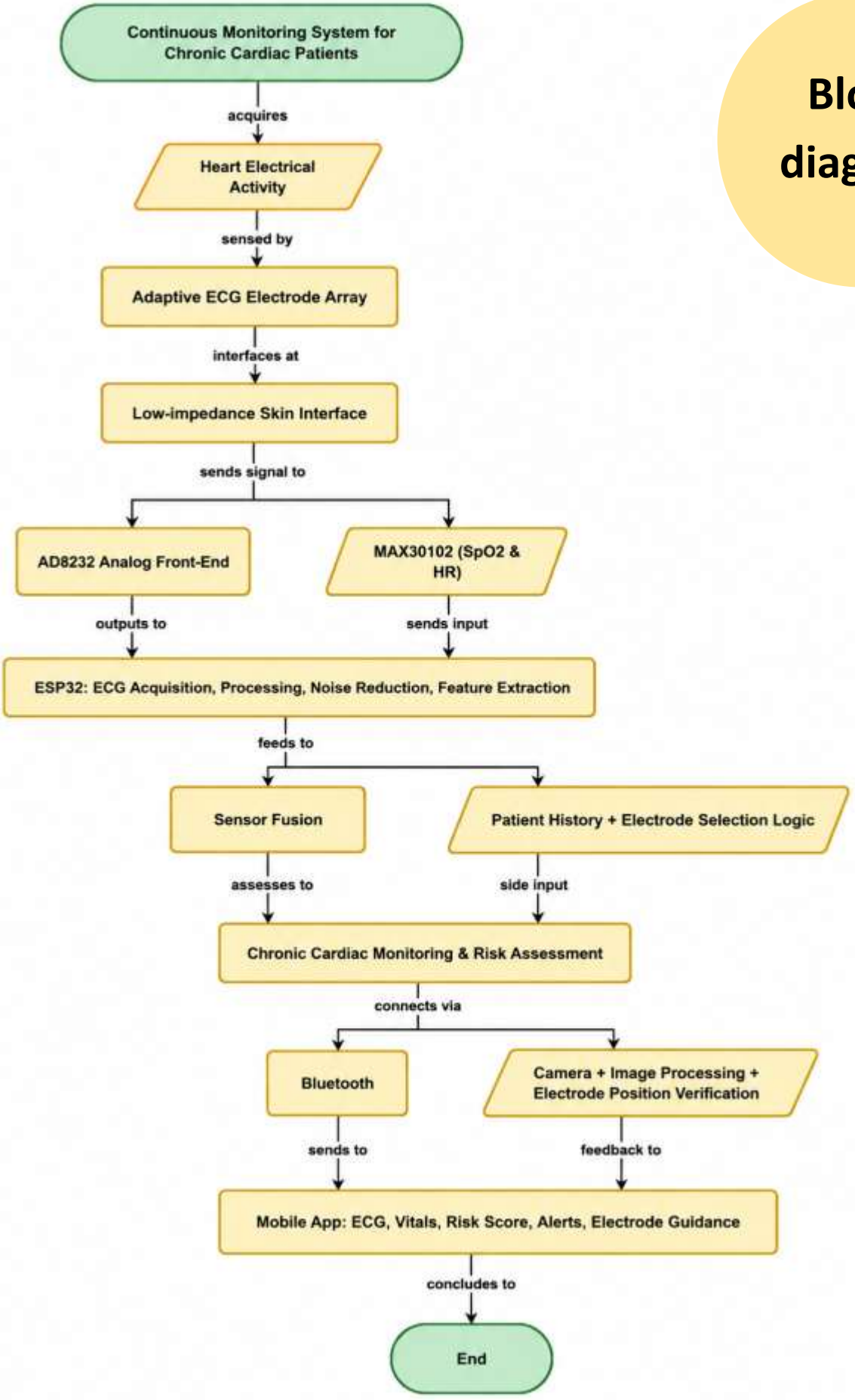
5. Communication Module

- Bluetooth BLE in microcontroller (ESP32)

6. Mobile Application

- ECG
- HR
- SpO_2
- Alerts
- Risk Score
- Electrode Guidance

Block diagram



Expected Impact

Clinical Impact

- Enable earlier recognition of myocardial ischemia and infarction through personalized ECG monitoring.
- Support timely medical intervention by detecting localized electrical abnormalities before severe clinical deterioration.
- Improve long-term follow-up for patients at high cardiovascular risk through continuous wearable monitoring.
- Reduce unnecessary emergency visits by providing continuous remote surveillance and early warning alerts.

Healthcare Impact

- Promote the shift from reactive healthcare to preventive healthcare, aligning with Egypt Vision 2030 and global digital health initiatives.
- Increase patient awareness and engagement through continuous monitoring and personalized feedback.
- Support telemedicine by enabling physicians to remotely review ECG trends and patient status.

Technical Impact

- Introduce a personalized electrode-placement strategy instead of a fixed wearable ECG configuration.
- Demonstrate the feasibility of combining targeted ECG acquisition with a lightweight four-electrode architecture.
- Provide a scalable platform that can integrate additional physiological sensors and AI-based diagnostic algorithms.

Economic & Social Impact

- Reduce healthcare costs associated with repeated hospital visits and delayed diagnosis.
- Improve accessibility to continuous cardiac monitoring for patients in remote and underserved communities.
- Enhance quality of life by enabling comfortable long-term monitoring outside healthcare facilities.

Expected Validation Outcomes

- Successful acquisition of ECG signals using the proposed four-electrode configuration.
- Stable wireless transmission of physiological data to the mobile application.
- Feasibility of patient-specific electrode placement based on clinical risk mapping.
- Demonstration, through simulation and literature-supported analysis, of the potential for improved localization of ischemic electrical changes compared with conventional fixed wearable configurations.
- Validation of the proposed system architecture as a foundation for future prototype development and clinical testing.

Future Work

Prototype Development

- Fabricate and experimentally evaluate the proposed conductive microneedle electrode.

- Integrate all hardware components into a compact wearable prototype.
- Optimize power consumption for long-term continuous operation.

Experimental Validation

- Compare signal quality with conventional Ag/AgCl electrodes.
- Evaluate skin–electrode impedance and motion artifact reduction.
- Assess the performance of the personalized electrode placement strategy across different cardiac conditions.

Clinical Validation

- Conduct pilot studies involving patients with varying cardiovascular risk profiles.
- Compare system performance with standard 12-lead ECG recordings under clinical supervision.
- Evaluate sensitivity and specificity for early detection of myocardial ischemia and infarction.

Artificial Intelligence

- Develop machine learning models for automated risk prediction and anomaly detection.
- Personalize electrode placement recommendations using patient-specific clinical data.

Product Development

- Miniaturize the hardware into a smartwatch-compatible or patch-based wearable platform.
- Develop a secure cloud infrastructure for remote physician monitoring.
- Pursue regulatory approval and commercialization after successful clinical validation.

Business model

BUSINESS MODEL CANVAS – PULSEFORGE

Continuous Cardiac Monitoring. Personalized Care. Smarter Outcomes.

PULSEFORGE
Smart ECG. Smarter Care.

KEY PARTNERS

- Universities and research centers
- Hospitals
- Ministry of Health
- Medical device manufacturers
- Graphene material suppliers
- Electronics manufacturers
- Healthcare investors
- Regulatory authorities

KEY ACTIVITIES

- Research and development
- Prototype manufacturing
- Clinical testing
- Regulatory certification
- Software development
- Marketing
- Production and assembly
- Distribution
- Customer support

KEY RESOURCES

- Graphene and PEDOT electrode technology
- Proprietary electrode design
- Embedded electronics
- AI software platform
- Clinical research partners
- Manufacturing facilities
- Intellectual property and patents

VALUE PROPOSITIONS

- Continuous real-time ECG monitoring
- Early detection of cardiac deterioration
- High-quality ECG signals with reduced motion artifacts
- Comfortable wearable chest binder suitable for long-term use
- Affordable compared with traditional Holter monitoring systems
- AI-ready platform for predictive cardiac analytics



CUSTOMER RELATIONSHIP

- Medical device training
- Technical support
- Mobile application for patients
- Physician dashboard for remote monitoring
- Subscription-based software updates

CHANNEL

- Direct sales to hospitals
- Medical equipment distributors
- Partnerships with healthcare organizations
- Online sales platform
- Government healthcare procurement
- Telemedicine platforms

CUSTOMER SEGMENTS

- Cardiology hospitals.
- Public and private healthcare providers.
- Home healthcare agencies.
- Telemedicine companies.
- Health insurance providers.
- Patients with chronic cardiovascular diseases.
- Elderly individuals and high-risk patients.

COST STRUCTURE

- Research and development
- Sensor manufacturing
- AI algorithm development
- Clinical validation
- Regulatory approval
- Marketing and sales
- Customer support
- Production and assembly
- Distribution
- Maintenance and infrastructure

REVENUE STREAMS

- Sale of wearable ECG devices
- Monthly subscription for cloud monitoring
- AI-powered ECG analysis subscription
- Hospital licensing agreements
- Maintenance and replacement electrode kits
- Data analytics services (with patient consent and regulatory compliance)

الموديل	حجم المبيعات التقريبي (سنوياً في مصر)	الإيرادات السنوية التقريبية (بالجنيه المصري)	العائد الاستثماري للأخصائي (ROI)
HeartRec Eco	600 - 800 جهاز	24,000,000 - 18,000,000 جم	**سريع جداً** (يسترد قيمته خلال 2-3 أشهر)
Contec TLC6000	700 - 900 جهاز	27,000,000 - 21,000,000 جم	**ممتاز** (معدل إهلاك منخفض وتكلفة صيانة شبه منعدمة)
Biocare iH 3 Plus	400 - 500 جهاز	15,000,000 - 12,000,000 جم	**مستقر** (جهاز يعيش طويلاً ومثالي للمرضى كبار السن)

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